WAWIRA JOSHUA NYAGA

TDC113-C002-0003/2024

COMPUTER SCIENCE–LEVEL 6
MATHEMATICS FOR COMPUTERS
1 1/2 HOURS
JULY - 2025

JKUAT TVET INSTITUTE – KAREN CAMPUS
WRITTEN ASSESSMENT
CAT 1

SECTION A (20 Marks). Answer all questions.

1. Solve. (4marks)

$$\frac{2x-1}{3}+\frac{x+2}{4}=3$$

2. The perimeter of a rectangle is 64 cm. The length is 4 cm more than twice the width. Form and solve an equation to find the dimensions of the rectangle. (3Marks)
3. An office secretary procured 3 printers and 1 computer at the cost of USD 700 in the first month. She later bought 5 printers and 2 computers at the cost of USD 1200 in the second month. Use the matrix method to determine the price of a printer and a computer. (5 marks)
4. Find the value matrix A, so that the equality is satisfied. (5 marks)

$$A+\begin{bmatrix}2 & 3\\ -4 & 1\end{bmatrix}=\begin{bmatrix}5 & -1\\ 1 & 5\end{bmatrix}$$

5. The scores obtained by 12 students in a test are
   6, 7, 6, 8, 9, 7, 6, 5, 8, 7, 6, 7
   Find the **mode**. (3Marks)

SECTION B (30 Marks) Answer any 3 Questions

6. Use the following grouped data to find the **mean**. (10Marks)

Class Interval	Frequency
0 – 10	5
10 – 20	7
20 – 30	12
30 – 40	6

7. The following is a frequency table of scores:

Class Interval	Frequency
0–9	2
10–19	5
20–29	9
30–39	7
40–49	4

Draw a **histogram** to represent this data.

(10Marks)

8. A car rental company charges a flat fee of KSh 1,000 plus KSh 20 per kilometre. Another company charges KSh 500 flat plus KSh 30 per kilometre. At what distance are the costs equal?

(10Marks)

9. A store sells two types of fruits: **apples** and **mangoes**.

- **2 apples and 3 mangoes cost KSh 90**
- **4 apples and 2 mangoes cost KSh 100**

Use matrices to find the cost of **one apple** and **one mango**.

(10Marks)

~~45/50~~ ~~45/50~~ ~~45/50~~ ~~35/50~~ ~~72%~~

JKUAT CONTINUOUS ASSESSEMENT TESTS

NAME WAKIIRA JOSHUA NYAGA

COURSE Computer Science YEAR OF STUDY 1

UNIT CODE ..... TITLE .....

REG NO TDC113-0002-0035/2024

SHEET NO .....

DATE 2/7/2025

NOTE: This stationery will be used for Continuous Assessment work only

It will be a breach of examination regulations to use it otherwise

1. $\frac{2x-1}{3} + \frac{x+2}{4} = 3$

2	3	4
2	3	2
3	3	1
	1	1

L.C.M $= 2^2 \times 3 = 12$

$\left(\frac{2x-1}{\cancel{3}}\right)^{\cancel{12}\,4} + \left(\frac{x+2}{\cancel{4}}\right)^{\cancel{12}\,3} = \cancel{3} \times 12$

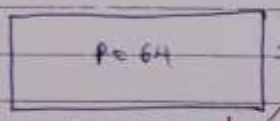
$4(2x-1) + 3(x+2) = 36$

$8x - 4 + 3x + 6 = 36$

$11x + 2 = 36$

$\frac{11x}{11} = \frac{34}{11}$ $x = 3\frac{1}{11}$

2.



$P = 2(L + W)$

$64 = 2(4 + 2x + x)$

$64 = 2(4 + 3x)$

$64 = 8 + 6x$

$\frac{6x}{6} = \frac{56}{6}$

$x = 9\frac{1}{3}$

Length $= 4 + 2(9\frac{1}{3})$

$4 + 18\frac{2}{3}$

$= 22\frac{2}{3}$

$22\frac{2}{3}$ by $9\frac{1}{3}$

3

07

3) Let tomatoes be x and computer be y

$2x + y = 700$

$5x + 3y = 1200$

$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 700 \\ 1200 \end{pmatrix}$

$d = (3\times2) - (1\times5)$

$6 - 5 = 1$

$\frac{1}{1} = 1 \quad 1\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$

$A^{-1}A = I$

$\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}\begin{pmatrix} 700 \\ 1200 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2\times700 + -1\times1200 \\ -5\times700 + 3\times1200 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1400 - 1200 \\ -3500 + 3600 \end{pmatrix}$

$x = 100$

$y = 100$

5

4) $A + \begin{pmatrix} 2 & 3 \\ -4 & 1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ 1 & 5 \end{pmatrix}$

$A = \begin{pmatrix} x & y \\ w & z \end{pmatrix}$

$\begin{pmatrix} x & y \\ w & z \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ 1 & 5 \end{pmatrix} - \begin{pmatrix} 2 & 3 \\ -4 & 1 \end{pmatrix}$

$\begin{pmatrix} x & y \\ w & z \end{pmatrix} = \begin{pmatrix} 3 & -4 \\ 5 & 4 \end{pmatrix}$

$A = \begin{pmatrix} 3 & -4 \\ 5 & 4 \end{pmatrix}$

5

5) 6, 7, 6, 8, 9, 7, 6, 5, 8, 7, 6, 7

mode is the number appearing many times

= 6 and 7

10

SECTION B

) Apples = x Mangoes = y

$2x + 3y = 90$

$4x + 2y = 100$

$$\begin{pmatrix} 2 & 3 \\ 4 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 90 \\ 100 \end{pmatrix}$$

$d = (2\times2) - (4\times3)$

$4 - 12 = -8$

$$\frac{1}{-8}\begin{pmatrix} 2 & -3 \\ -4 & 2 \end{pmatrix} = \begin{pmatrix} -1/4 & 3/8 \\ 1/2 & -1/4 \end{pmatrix}$$

$A^{-1}A = I$

$$\begin{pmatrix} -1/4 & 3/8 \\ 1/2 & -1/4 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 4 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1/4 & 3/8 \\ 1/2 & -1/4 \end{pmatrix}\begin{pmatrix} 90 \\ 100 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1/4 \times 90 + 3/8 \times 100 \\ 1/2 \times 90 + -1/4 \times 100 \end{pmatrix}$$

10

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -22\frac{1}{2} + 37\frac{1}{2} \\ 45 + -25 \end{pmatrix}$$

$x = 15$ One Apple = Sh. 15

$y = 20$ One Mangoe = Sh. 20

6) Class interval

	f	x	fx	Tally
0-10	5	5	25	~~||||~~
10-20	7	15	105	~~||||~~ ||
20-30	12	25	300	~~||||~~ ~~||||~~ ||
30-40	6	35	210	~~||||~~	

$\Sigma f = 30$ $\Sigma fx = 640$

10

$$\text{mean} = \frac{\Sigma fx}{\Sigma f} = \frac{640}{30}$$ mean = 21.33

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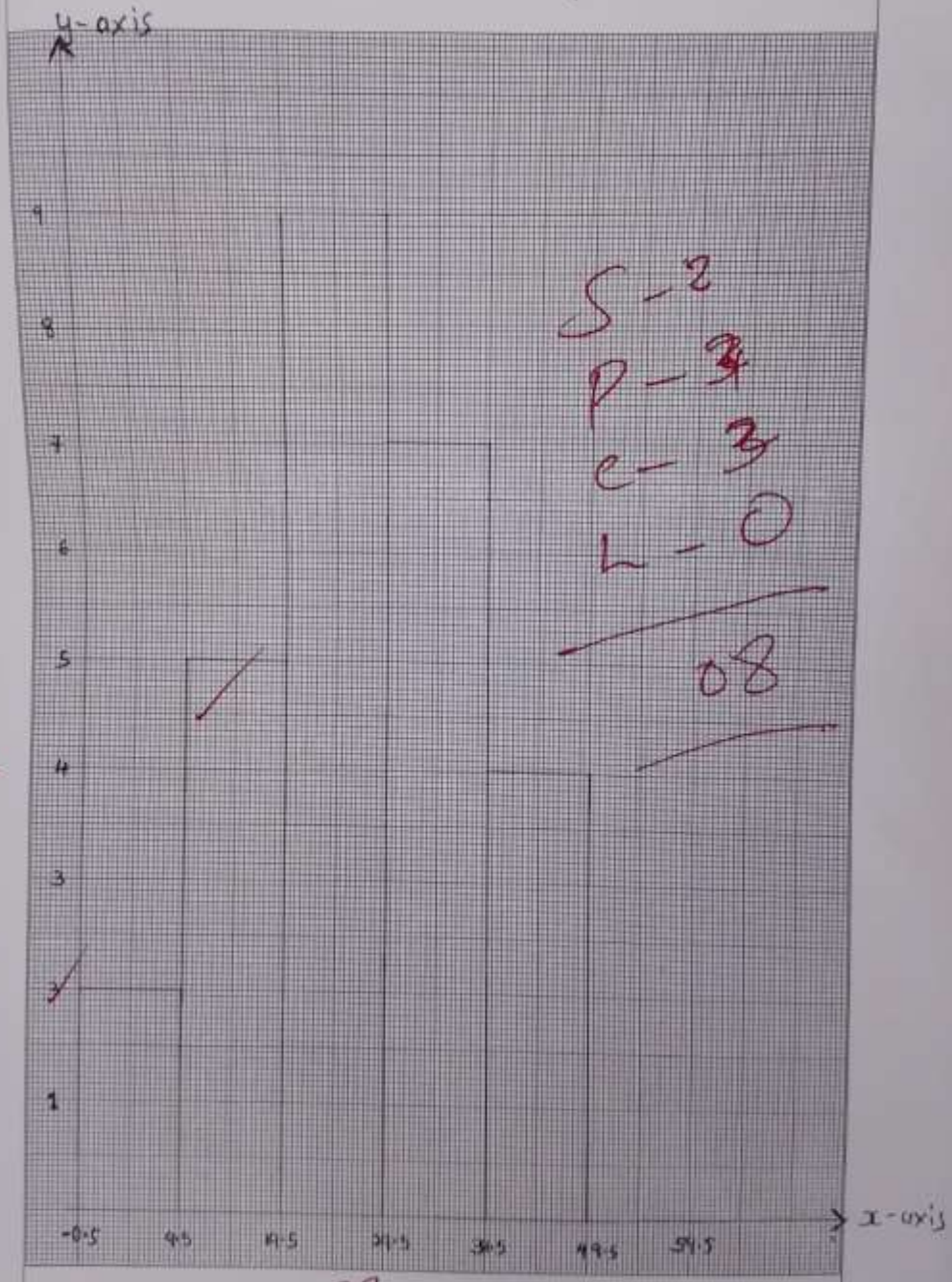

S - 2
P - 3
C - 3
L - 0
08

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